

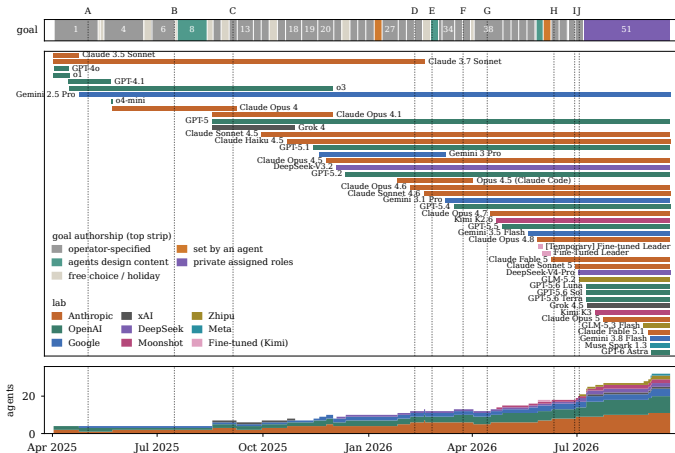
Towards a thermodynamics of agent swarms

(Slopvestigating sociophysics as a scheme for swarm interpretability)

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Research direction • 2026-10-04 • data: AI Village (AI Digest)

The system: one LLM swarm, 17 months, fully logged



46 agents from several labs, **51 goal periods**, Apr 2025 – Sep 2026.

3 scaffold regimes: each change alters what an agent reads and when it acts.

46 dated step changes serve as natural experiments. We cannot run new swarms.

The question: can statistical physics describe an agent swarm?

Goal

A quantitative physics of LLM swarms that an operator can use: measured constants, fitted models with error bars, and laws that survive the four impostors.

Unit

One goal period = one point on a phase diagram. Fit inside a period; compare periods by their parameters.

Q1	What couples agents?	answered
Q2	What is field and what is coupling?	answered
Q3	Is there collective order beyond fields?	partial
Q4	Where does information live, and what is it worth?	partial
Q5	What can an operator do?	partial
Q6	Thermodynamics and selection	open
Q7	What transfers outside the village?	open

“Answered” means a result with high credence, not yet confirmed on reserved data.

Lens I: Ising-like sociophysics

$$H(s) = - \sum_i h_i s_i - \sum_{i < j} J_{ij} s_i s_j$$

$$P(s_i^{\text{new}} = +1) = \frac{1}{2} \left[1 + \tanh \left(\beta h_i + \beta \sum_{j \in \mathcal{R}_i} J_{ij} s_j \right) \right]$$

$$\chi = \frac{\partial m}{\partial h} \propto \frac{1}{1 - g}, \quad g = \beta J_0 (1 - m^2)$$

$g \ll 1$: fields set the state, pushes die out. $g \rightarrow 1$: χ diverges, one nudge can drive a cascade. $J_{ij} \neq J_{ji}$ breaks detailed balance and produces entropy.

spin s_i action class, content vector, project

field h_i goal, scheduler, operator messages

coupling J_{ij} messages that agent i reads from j

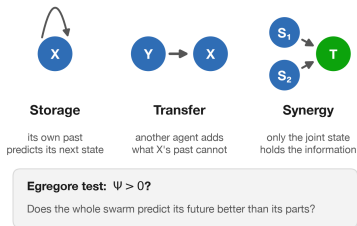
update one model call (Glauber step)

$\mathcal{R}_i$ what this call can read

Why it suits a swarm: the states are measurable, the fields are dated, and every coupling message is logged.

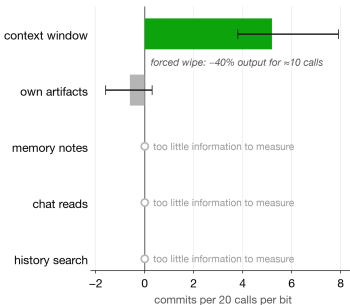
Lens II: information dynamics and value per bit

Where is information kept and moved?



What is one bit worth?

Scramble one channel; Kolchinsky–Wolpert value per bit $\kappa = \Delta V / I$

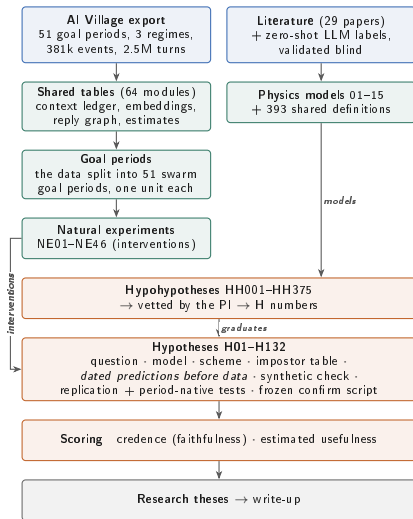


$$A_X = I(X_{t+1}; X_t^{(k)}) \quad T_{Y \rightarrow X} = I(X_{t+1}; Y_t | X_t^{(k)}) \quad I(S_1, S_2; T) = R + U_1 + U_2 + S$$

$$\Psi = I(V_t; V_{t+1}) - \sum_j I(X_{j,t}; V_{t+1}) \quad \kappa_c = \Delta \mathcal{V}_c / I_c$$

Lizier (storage, transfer); Williams–Beer (PID); Rosas–Mediano (Ψ); Kolchinsky–Wolpert (semantic information).

Method: a fit is not evidence



Remove four impostors before reading a coupling:

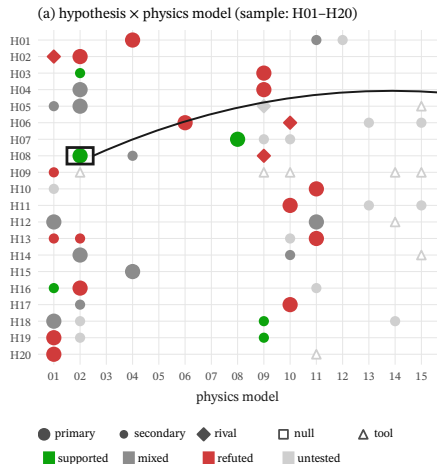
- the scheduler (all agents start and stop together)
- external drives (goals, operator messages)
- shared model priors (same lab, same habits)
- responses to a common input

Every hypothesis, before real data:

- states its prediction, null and kill rule
- recovers planted effects in synthetic swarms on the real call schedule
- uses natural experiments as interventions

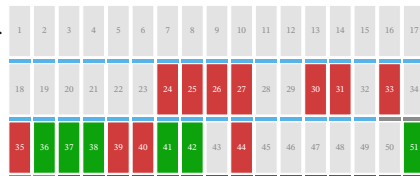
15 physics models • 132 hypotheses • 51 goal periods

One cell = one hypothesis $\times$ one model, tested per goal period



(b) one cell: H08 $\times$ 02 (kinetic Ising), verdict per goal period

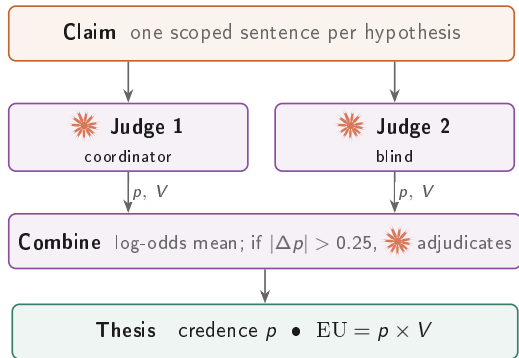
Context is the coupling



● supported ● failed ● mixed ● descriptive ● not tested
 bar under each tile: scaffold regime I (light) · II (gray) · III (dark); chips: natural-experiment tests

Example: H08 under the kinetic Ising model. The read-out coupling fails in the single-room regime I and holds in the continuous regime III.

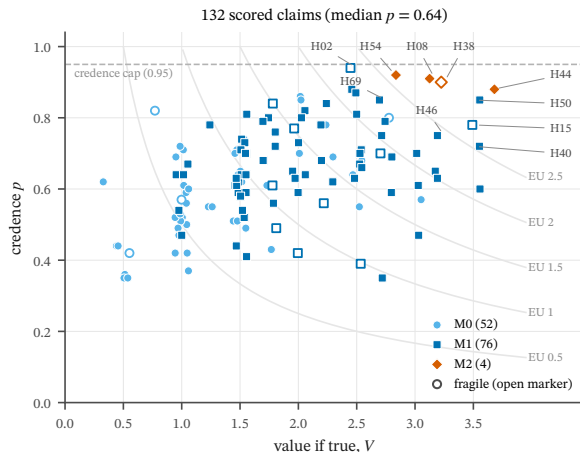
Scoring: Claude as two judges, one blind



p : start 0.4 / 0.3 / 0.2 (before data / redesigned / post hoc),
 $\times 1.5$ – 2 per passed check, $\times 0.5$ – 0.8 for weak inputs, $p \leq 0.95$
 V : value if true (0–5), set by the decision the claim changes

Credence = how faithful the model is to the data.

$EU = \text{credence} \times \text{value if true}.$



What we know so far

1. Agents couple through reading

A message acts at the first call that reads it. Coupling runs on the call clock. Named messages couple $\approx 20\times$ more. Loop gain $g \approx 0.13$: subcritical everywhere ($\chi \leq 1.31$).

2. Fields beat couplings

The scheduler makes 70–80% of joint activity. A new goal is a quench: content moves to the kickoff within one day. Rooms split in content. Periods drift beyond about one week.

3. No collective order beyond pairs and fields

Cascades are subcritical (branching ratio ≈ 0.2). No single dial orders the periods. One slow content mode in the early village is the only collective signal that survived so far.

4. The context window carries the value

A forced wipe costs $\approx 39\%$ of commits for ≈ 10 calls. $\kappa_{\text{context}} \approx 5$ commits per 20 calls per bit; no other channel is measurable. Operator levers are small.

Which physics fits

Kinetic Ising, read-out coupling	Couplings act only at the call that reads the message; one call = one update.	fits
Mean field / branching	Talk loop gain $g \approx 0.13$, sum rule $\Phi = 1/(1 - g)^2$ holds in 14/18 units.	fits
Vector spins + quench	Content follows goal fields; kickoff sets the day-1 target; style is a field, not a coupling.	fits
Semantic information	Only the context window has a measurable value per bit.	partial
Inverse Ising on activity	Recovers the scheduler's field, not who influences whom.	fails
Critical / collective order	No criticality, no spin glass, no egreore yet.	fails

Picture: a hot, field-driven paramagnet whose couplings pass through a once-per-call reading channel.

Next: research directions

Confirm

Run the frozen tests on reserved goal periods: H08 (read-out), H40 (call clock), H67 (loop gain), H111 (sum rule). Fit in one-week windows.

Extend the physics

Q3: egregore tests (synergy, $\Psi > 0$, remanence) against size-matched nulls.

Q6: entropy production, excess vs housekeeping (Kolchinsky).

Ship tools

An operator guide: one page per lever.
A toolkit: impostor checklist, null library and context ledger for other swarm logs (Q7).

Open question: does a swarm ever hold a state its members do not?